Quantum Many-Body Models: Characterization & Method Selection

A property checklist and the methods it gates — complexity, applicability, tasks

This report characterizes a quantum many-body lattice model by **16 property axes** (A1–D16) and maps those axes to the numerical methods they enable or block, including algorithmic complexity expressed in the same property metrics. Companion sources: `model-property-checklist.md`, `method-property-map.md`, `method-survey.md`. Complexity exponents were web-verified (2026-06) against the primary literature.

Cost symbols. N sites/orbitals · d local dim · $D_H = d^N$ Hilbert dim · D_{blk} largest symmetry block · χ MPS bond · D PEPS bond · χ_{env} environment bond · M Chebyshev moments · N_s samples · N_w walkers · β inverse temp · $N_\tau = \beta/\Delta\tau$ · τ_{ac} autocorr. time · z dyn. exponent · $\langle s\rangle$ avg. sign · n qubits · N_p params · N_c cluster size · D_c thermal bond.

1. Model characterization — the 16 property axes

Axis	Possible values	Why it matters (method gating)
A · Geometry & Hilbert space		
A1 Dimensionality & geometry	1D / quasi-1D / 2D / 3D / ∞-D; lattice; coordination Z	Master gate — sets entanglement scaling and fluctuation strength, hence the method family.
A2 Boundary conditions	OBC / PBC / cylinder / torus / infinite	Finite-size effects, cut entanglement, access to topological diagnostics.
A3 Statistics & local dim d	spin / boson / fermion / hard-core / anyon; d	Fermions/anyons → sign; d sets the d^N wall and per-site TN cost.
A4 Interaction range	short-range / long-range ($1/r^\alpha$, Coulomb)	Long-range can violate the area law; needs many-term MPOs.
B · Phase & entanglement		
B5 Entanglement scaling	area / area+log (1D crit.) / volume; TEE	Decides TN feasibility via $\chi \gtrsim e^S$; volume law breaks TN.
B6 Spectral gap	gapped (finite ξ) / gapless–critical	Gapless inflates χ and worsens finite-size scaling.
B7 Ground-state order type	trivial / SSB / SPT / topological / spin liquid	Sets degeneracy assumptions and diagnostics (order param, ES, MES+TEE).
B8 Frustration	none / geometric / interaction / fermionic	Triggers the QMC sign problem; inflates TN χ.
C · Symmetry & solvability		
C9 Global / internal symmetry	U(1) / SU(2) / Z_2 / particle-hole / TRS	Block-diagonalizes H — cheapest cost cut; PH → sign-free at half-filling.
C10 Spatial / lattice symmetry	translation (k) / point group / inversion	$\sim N\times$ ED reduction; irrep labels for excited states.
C11 Integrability	free-fermion / Bethe-ansatz / non-integrable	Quadratic → $O(N^3)$ exact; analytic benchmarks; forbids thermalization.
C12 Sign problem	sign-free / mild / severe	Gates QMC: $\langle s\rangle \sim e^{-\beta N \Delta f}$; generically NP-hard.
D · Regime & complications		
D13 Computational regime	ground state / finite-T / real-time	Real-time → $\chi \sim e^t$; finite-T → LTRG/METTS/QMC.
D14 Filling / doping	commensurate / incommensurate–doped	Doping breaks PH symmetry → turns on the fermion sign problem.
D15 Disorder / MBL	clean / disordered; ergodic vs MBL	× realizations; MBL keeps excited-state area law (log-t growth).
D16 Hermiticity	Hermitian / non-Hermitian–open	Non-Hermitian breaks the variational principle; needs density-matrix/trajectory methods.

2. Method family overview — "which method when"

Method	Dim	Reachable size	Regime	Hard blocker
ED	any	tiny ($d^N \lesssim 10^8$)	GS + full spectrum + dynamics	Hilbert space d^N
MPS / DMRG	1D, quasi-2D	large 1D / narrow cylinder	GS, finite-T, dynamics	1D area law (χ)
PEPS	2D	moderate	GS (finite-T harder)	#P-hard contraction
QMC	any	large	GS + finite-T	sign problem

Method	Dim	Reachable size	Regime	Hard blocker
VMC / NQS	any	large	GS (+dynamics)	variational / ansatz bias
MF / HF	any	unlimited	baseline	neglects correlation
LTRG / XTRG	1D/2D	large	finite-T thermodynamics	ground state / 3D
MCRG	any (classical)	large near T_c	critical exponents	non-critical / GS
Circuit sim	—	n qubits (2^n mem)	variational circuits / VQE	state-vector memory
PolyOpt (SDP)	1D/2D	~100 (1D) / 10×10 (2D)	certified GS bounds	3D / SDP blow-up
DMFT	high / ∞-D	infinite (local)	finite-T, Mott, dynamics	neglects spatial corr.

3. Algorithmic complexity & its driving property

The driving-property column names the axis (A1–D16) that dominates the scaling — e.g. ED is gated by symmetry (C9/C10), QMC by the sign problem (C12), tensor networks by entanglement (B5).

Method	Time (leading)	Memory	Driving property metric
Exact & classical Monte Carlo			
ED — full	$O(D_{\text{blk}}^3)$	$O(D_{\text{blk}}^2)$	A3 d^N, reduced by C9/C10
ED — Lanczos	$O(n_{\text{it}} N D_{\text{blk}})$	$O(D_{\text{blk}})$	C9/C10; B6 $\to n_{\text{it}}$
FTLM / TPQ	$O(R n_{\text{it}} N D_{\text{blk}})$	$O(D_{\text{blk}})$	C9/C10; D13
KPM	$O(MRN_{\text{nnz}})$	$O(D_H)$	A4 sparsity; M = resolution
Classical MC (local)	$O(N\tau_{\text{ac}}N_s)$	$O(N)$	B6 $\to \tau_{\text{ac}} \sim L^z$ $(z \approx 2)$
Cluster MC	$\approx O(NN_s)$ near T_c	$O(N)$	B8 (breaks clusters)
MCRG	MC × RG steps	$O(N)$	B6
Tensor networks			
DMRG / MPS	$O(Nwd\chi^3)$	$O(d\chi^2)$	B5/B6/B8 $\to \chi$; A1 width $\to e^{cW}$; C9 ↓
iPEPS	$O(\chi_{\text{env}}^3 D^6)$ to $O(D^{12})$	$O(\chi_{\text{env}}^2 D^4)$	B5/B8 $\to D$; full-update $O(D^{10})$
MERA	$O(\chi^7)$ to $O(\chi^9)$	$O(\chi^4)$	B6 criticality $\to \chi$
TRG / HOTRG	$O(\chi^6)$ / $O(\chi^{4d-1})$	$O(\chi^4)$ / $O(\chi^{2d})$	B6 $\to \chi$; A1 dim $\to$ exponent
TEBD / TDVP	$O(Nwd\chi(t)^3)$	$O(d\chi^2)$	D13+B5 $\to \chi(t) \sim e^{S(t)}$; A4 $\to w$
purification / METTS	$O(Nd^2\chi^3)$	$O(d^2\chi^2)$	D13; low-T $\to \chi$
LTRG / XTRG	poly(d, D_c) × $(\beta/\tau$ or $\log\beta)$	$O(D_c^2)$	D13; low-T $\to D_c$; A1 (2D ok)
Quantum Monte Carlo			
SSE / worldline	$O(N\beta/\langle s\rangle^2)$	$O(N\beta)$	C12 $\langle s\rangle$; B6 $\to$ updates
DQMC	$O(\beta N^3/\langle s\rangle^2)$	$O(N^2N_\tau)$	C12 $\langle s\rangle \sim e^{-\beta N\Delta f}$
AFQMC / CPMC	$O(N^3N_wN_{\text{step}})$	$O(N^2N_w)$	C12 (constraint bias)
VMC / NQS	$O(N_sc_{\text{ev}} + N_p^2)$	$O(N_p)$	ansatz quality (not C12)
DMC / GFMC	$O(N_wN_{\text{proj}}c_{\text{step}})$	$O(N_wN)$	C12 (fixed-node bias)
DiagMC	grows w/ diagram order	—	C12 + series convergence
Embedding, perturbative, mean-field, circuit, certified			
DMFT (+solver)	loop × solver; cluster $\sim e^{N_c}$	solver	A1 Z; C12 (solver); N_c
DMET	solver on $2N_{\text{frag}}$	solver	fragment size
NRG	$O\left(N_{\text{kept}}^3\right)$ / iter	$O\left(N_{\text{kept}}^2\right)$	A3 channels $\to N_{\text{kept}}$
fRG	$\sim O\left(N_{\text{patch}}^3\right)$	$O\left(N_{\text{patch}}^2\right)$	coupling strength (truncation)
CCSD / CCSD(T)	$O(N^6)$ / $O(N^7)$	$O(N^4)$	B7 single-ref validity; N orbitals
GW / RPA / GF2	$O(N^4)$ / $O(N^4)$ / $O(N^5)$	$O(N^3)$	weak coupling; N orbitals
HF / DFT	$O(N^{3-4})$ / iter	$O(N^2)$	A1 Z (accuracy; corr. $\sim 1/Z$)
Spin-wave (1/S)	$O(m^3)$ × k-points	$O(m^2)$	B7 order required; A3 large-S
Circuit — state-vector	$O(\text{gates}\cdot 2^n)$	$O(2^n)$	n
Circuit — tensor network	$O(\exp(\text{treewidth}))$	varies	B5 / depth $\to$ treewidth
Circuit — stabilizer	$O(n)$/gate, $O(n^2)$/meas.	$O(n^2)$	exp. in non-Clifford gates
SDP / NCTSSOS	high-poly in $O(N^k)$	$O(N^{2k})$	level k; C9 ↓; A1 dim

4. Property → method gate

Property value	Favors	Blocks / expensive
A1 = 1D	MPS (near-exact), ED	PEPS overkill
A1 = 2D	PEPS, sign-free QMC, DMRG cylinders	DMRG e^{cW} wall
A1 = 3D / ∞-D	QMC, (cluster-)DMFT (∞-D exact)	DMRG / PEPS impractical
A3 fermions	AFQMC, DMRG (JW), DMFT	QMC sign; TN parity bookkeeping
A4 long-range	TDVP, MPO W^{II}; ED	TEBD; denser ED matrix
B5 area law	MPS, PEPS	—
B5 volume law	ED (small), QMC (eq. finite-T), VMC	tensor networks fail
B6 gapless / critical	MERA, MCRG, TN at large χ	MF (false transitions)
B7 topological order	DMRG (MES protocol), ED on torus	unique-GS methods alone
B8 frustration	DMRG, VMC/NQS, ED, PolyOpt	QMC (sign blocked)
C9 / C10 symmetry	ED & DMRG (large speedup), smaller SDP	—
C11 integrable	exact $O(N^3)$ / Bethe / TBA	(numerics only validates)
C12 sign-free	QMC (exact at large size)	—
C12 sign-ful	MPS / PEPS / VMC, PolyOpt bound	QMC
D13 finite-T	LTRG/XTRG, thermal QMC, METTS, DMFT	—
D13 real-time	TEBD/TDVP (short t), ED-Krylov	QMC (dynamical sign)
D16 non-Hermitian	density-matrix / trajectory methods	variational GS methods

5. Tasks → methods (and how cost scales)

Task	Methods (cheap → expensive, within reach)	Cost note
Ground-state energy	integrable-exact ≪ MF ≪ DMRG(1D) / sign-free QMC / AFQMC / VMC ≪ iPEPS(2D) ≪ ED	TN cost ← B5; QMC ← C12
Full spectrum / level statistics	ED (full) only	$O(D_{\mathrm{blk}}^3)$; needs C9/C10
Few excited states / gap	ED Lanczos, DMRG (targeted)	per-state ≈ GS cost
Finite-T thermodynamics	classical/quantum MC, LTRG/XTRG, FTLM/TPQ, DMFT, purification/METTS	low-T raises χ / sign
Real-time dynamics	ED-Krylov (small), TEBD/TDVP, t-VMC	entanglement barrier; QMC blocked
Spectral function $S(q,\omega)$	KPM, ED continued-fraction, DMFT, DQMC + anal. cont., TEBD+FT	analytic continuation ill-posed
Critical exponents / CFT data	MCRG, TRG/TNR, MERA, finite-size scaling	needs B6 criticality
Order parameters / phase diagram	MF, QMC, DMRG/iPEPS, VCA, DMFT	accuracy vs method bias
Entanglement entropy / spectrum	DMRG/MPS (free), replica-trick QMC, ED	SPT/topo diagnostics (B7)
Topological order (TEE, degeneracy)	DMRG (MES), ED on torus	needs A2 torus/cylinder
Rigorous energy bracket	VMC (upper) + SDP/NCTSSOS (lower)	two-sided; works in C12 sign-ful
Spectra of correlated solids	DMFT/CDMFT/DCA, GW+DMFT, NRG (impurity)	momentum needs clusters

6. Per-model worked examples — the checklist applied

Each column fills in the 16 axes for a canonical harness model; the bottom row reads off the recommended method(s). Contrast the solvable cases (Heisenberg chain, TFIM) with the hard frontiers (kagome, doped Hubbard).

Axis	Heisenberg chain (1D, S=1/2)	Heisenberg kagome	Hubbard square (doped)	TFIM (1D)	Anderson impurity (symmetric)
A · Geometry & Hilbert space					
A1 dim / geom	1D chain, $Z = 2$	2D kagome, $Z = 4$	2D square, $Z = 4$	1D chain, $Z = 2$	0D impurity + bath
A2 boundary	OBC / PBC	cylinder / torus	cylinder / torus	any	Wilson chain / finite bath
A3 stat. & d	spin-1/2, $d = 2$	spin-1/2, $d = 2$	fermion, $d = 4$	spin-1/2, $d = 2$	fermion, $d = 4$
A4 range	NN	NN	NN (t, t')	NN	impurity–bath hybridization
B · Phase & entanglement					
B5 entanglement	area+log ($c = 1$)	2D area law; TEE candidate	2D area law	area (+log at QCP, $c = 1/2$)	low (impurity); bath area-law
B6 gap	gapless	contested (gapped vs gapless)	metallic / pseudogap	tunable; gapless at $\Gamma = J$	T_K (exp. small)
B7 order	algebraic, no SSB	spin-liquid candidate	competing stripe / SC / AFM	Z_2 SSB ↔ paramagnet	Kondo singlet / local moment
B8 frustration	none	geometric (strong)	fermionic (doping)	none	none
C · Symmetry & solvability					
C9 global sym	SU(2)	SU(2)	U(1)×U(1), SU(2)	Z_2	U(1)+SU(2), PH
C10 spatial sym	transl., inversion	transl. + C_{6v}	transl. + C_{4v}	transl., inversion	— (0D)
C11 integrability	Bethe-ansatz	non-integrable	non-integrable	free-fermion (JW)	Bethe-ansatz (Kondo)
C12 sign problem	sign-free (Marshall)	sign problem	sign problem (doped)	sign-free	sign-free
D · Regime & complications					
D13 regime	GS / any	GS	GS / finite-T	any	GS / finite-T / dynamics
D14 filling	— (spin)	— (spin)	doped (incommensurate)	— (spin)	half (symmetric)
D15 disorder	clean	clean	clean	clean	clean
D16 hermiticity	Hermitian	Hermitian	Hermitian	Hermitian	Hermitian
→ Recommended	DMRG (trivial), Bethe-ansatz exact, ED; sign-free QMC	DMRG cylinders + VMC/NQS + ED; PolyOpt bound; **QMC blocked**	DMRG cylinders, AFQMC (constrained), VMC/NQS, DMET, cluster-DMFT	exact Bogoliubov $O(N^3)$; DMRG/TEBD; sign-free QMC; ED	NRG (gold standard), ED (finite bath), CT-QMC, DMRG (Wilson chain)

7. Harness method skills

The harness's `method-*` skills, each a route-and-tool selector that hands off to a tool skill under `skills/`. Scope (what it can solve), computed properties, leading time complexity, and trade-offs.

Skill → tool	Scope / models (input)	Computes (properties)	Time complexity	Pros / cons
`/method-ed` → xdiag, quspin	any finite cluster (spin/fermion/boson), all symmetry sectors; $N \lesssim 40$–50 (spin-1/2)	full spectrum, GS & excited states, gaps, level statistics, scars/ETH, $S(q,\omega)$, finite-T (FTLM/TPQ), entanglement	full $O(D_{\text{blk}}^3)$; Lanczos $O(n_{\text{it}} N D_{\text{blk}})$	1. exact, no sign problem, any model, all observables − exponential wall, finite cluster only
`/method-mps` → mpskit, tenpy, itensors	1D & quasi-1D (narrow cylinder); finite (DMRG/TEBD) & infinite (VUMPS/ iDMRG/TDVP)	GS energy, order params, correlations, gaps, entanglement spectrum, dynamics, finite-T	$O(Nwd\chi^3)$; 2D cyl. $\chi \sim e^{cW}$	1. near-exact in 1D, area-law optimal, thermodynamic limit − 2D wall, volume-law fails
`/method-peps` → pepskit	2D quantum lattices (GS) + classical partition functions; iPEPS infinite; fermionic	GS energy, order params, correlation length, free energy	contraction $O(\chi_{\text{env}}^3 D^6)$; full update $O(D^{10})$	1. native 2D area law, frustration OK, thermo. limit − #P-hard contraction, steep D, dynamics hard
`/method-qmc` → sse, cpmc-lab	SSE: sign-free spin/boson finite-T. CPMC/AFQMC: single-band Hubbard GS (CPMC-Lab, pedagogical)	finite-T thermodynamics, structure factors, stiffness; fermion GS energy & correlations	SSE $O(N\beta)$; DQMC $O(\beta N^3)$	1. any dimension, large sizes, exact when sign-free − sign problem (frustration / doping / real-time)
`/method-vmc` → netket, jax	frustrated / 2D / sign-ful regimes; ansatz benchmarks, NQS training (needs `make install netket`)	variational GS energy (upper bound), observables, some dynamics	$O(N_s c_{\text{ev}} + N_p^2)$ per step	1. no sign problem, frustration OK, expressive NQS − ansatz bias, non-convex, upper bound only
`/method-mf` → Julia SCF (no tool skill)	lattice fermions (HF/UHF), spin models (Weiss decoupling); any dimension	order parameters, phase diagrams, mean-field bands	$O(N^{3-4})$ / SCF iter	1. fast baseline, any dimension, seeds correlated methods − no correlation/entanglement, fails in 1D & frustration
`/method-ltrg` → itensors (from scratch)	finite-T thermodynamics of 1D/quasi-1D & 2D quantum lattices; sign-free in 2D	free energy, internal energy, specific heat, susceptibility	poly$(d, D_c) \times (\beta/\tau)$; XTRG $O(\log\beta)$	1. finite-T in 2D without sign problem, frustration OK − GS / 3D out, low-T error accrues, no dynamics
`/method-mcrg` → jax (from scratch)	classical lattice spin models near criticality; any dimension; quenched-disorder variant	critical exponents $(y_t, y_h \to \nu, \beta, \gamma)$, RG flow / fixed point	MC sampling × RG iterations	1. direct exponents, large lattices (no critical slowing) − critical-point only, not GS / thermo observables
`/method-qcs` → tensorcircuit-ng, jax	parameterized circuits / VQE; differentiable circuit simulation; H as dense / sparse / MPO	circuit expectation values, VQE energy, gradients, performance profiles	state-vector $O(\text{gates} \cdot 2^n)$; TN $O(\exp(\text{tw}))$	1. differentiable, GPU, MPS/TN backends for low entanglement − 2^n memory wall, classical simulation only
`/method-polyopt` → qmbcertify, nctssos	certified GS lower bounds (1D/2D Heisenberg via qmbcertify; any algebra via nctssos), Bell, state-polynomial	rigorous energy lower bound, certified observable bounds	SDP; moment matrix $O(N^k)$	1. rigorous (complements VMC upper bound), no sign problem, symmetry-reducible − no wavefunction, 3D blows up, finite-size

Provenance. Axes A1–D16 from `model-property-checklist.md`; gating and complexity from `method-property-map.md` and `method-survey.md`. Exponents web-verified 2026-06 (Schollwöck 2011; Sandvik; Becca–Sorella; Troyer–Wiese; Xie et al. HOTRG; Wietek–Läuchli; Georges et al. DMFT; Aaronson–Gottesman; Navascués–Pironio–Acín / Wang et al. NCTSSOS). Key corrections folded in: HOTRG $O(\chi^{4d-1})$; MF corrections $\sim 1/Z$; ED frontier 48–50 sites.